Stitching Program Report

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CSC 340
04/21/2025

Main Steps

The program begins by selecting two images for stitching and loading them in grayscale, then uses SIFT to detect keypoints. These key points are matched using OpenCV's matcher with a ratio test to get rid of bad matches. After getting the good matches, the program creates a homography matrix using the RANSAC algorithm. This is done by selecting random subsets of matches, creating a homography, and iterating many times to minimize the total distance error. The program then compares the homography matrix made by the my RANSAC algorithm with the one made using OpenCV's function.

Approach Taken

The program follows the outline in the assignment but uses the RANSAC algorithm for calculating the homography matrix instead of using OpenCV's built in RANSAC function for homography. This helps understand how RANSAC works and how the homography matrix is created from a set of points. Although OpenCV can be used to compute the homography matrix, the assignment needs a manual approach to help better understand the algorithm.

RANSAC Performance

The number of iterations in RANSAC is important for getting an accurate homography. With fewer iterations, the result may not be correct, and the matrix values could be skewed. Increasing the number of iterations, in the range of 20,000 to 50,000, helps reduce the error, which gives a more stable result. The matrices computed by RANSAC and OpenCV are not exactly the same, but their diagonal values and the right column are close and are in a similar range, meaning that both methods are getting the same transformation.

Most Difficult Component

The most challenging part of the assignment was implementing the RANSAC algorithm for computing the homography matrix. Random sampling, constructing the homography matrix, and making sure the algorithm gives the correct result after many iterations required some trial and error. Also, managing the division by zero issue and minimizing the error with each iteration was a bit tricky.